



Who's On First: Arrival Date and Song Speed in Chipping Sparrows

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INTRODUCTION

Among migratory birds, such as American Redstarts, earlier arrivals are found to be in better physical condition, getting the first pick of the territory and thus having higher reproductive fitness [2]. Research also shows that among another migratory bird, the Chipping Sparrow, trill rate can serve as an indicator of territorial threat, with individuals responding more aggressively to those with higher trill rates [1]. We sought to bridge the gap between these studies and investigate the relationship between arrival time and song speed/trill rate.

We hypothesized that Chipping Sparrows who arrive to Western Massachusetts earlier will perform faster songs than those who arrive later.

METHODS & MATERIALS

- Study species: Chipping Sparrow (*Spizella passerina*)
- Surveys conducted along predetermined tracks around UMass Amherst Campus every morning, recording audio of individuals' songs from late March to early May
- Arrival date regarded as the first day an individual was reported singing, determined via individual-specific songs [Fig. 1] and the birds' territory location
- Identified individuals' song length and number of beats per song using Raven audio analysis software [Fig. 2]
- Song speed: beats per second
- Linear regression between arrival date, measured in days since April 1st, and song speed run in Statdisk statistical analysis software

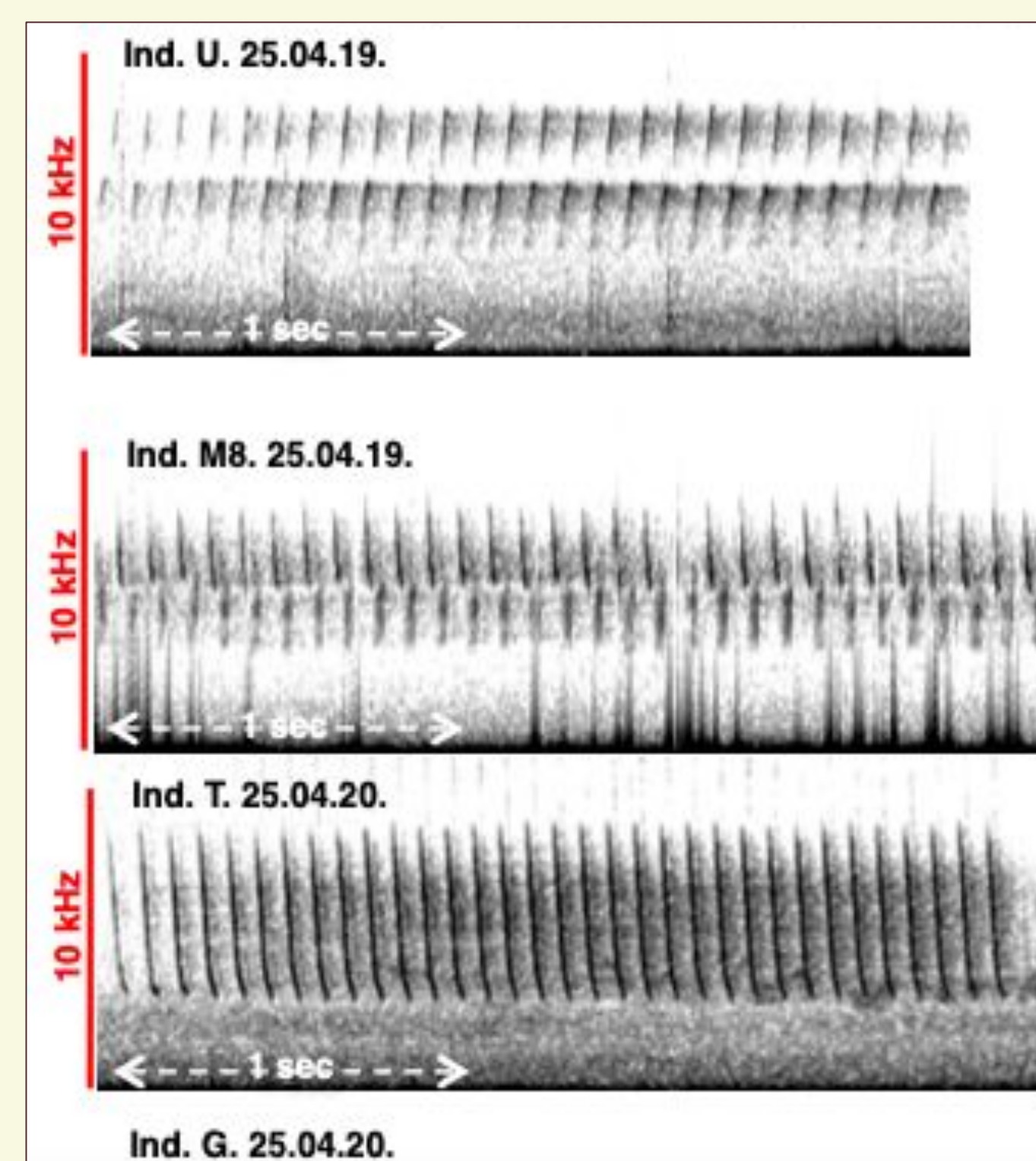


Figure 1: Examples of individual song variations [Dr. Ju-Hyun Lee]

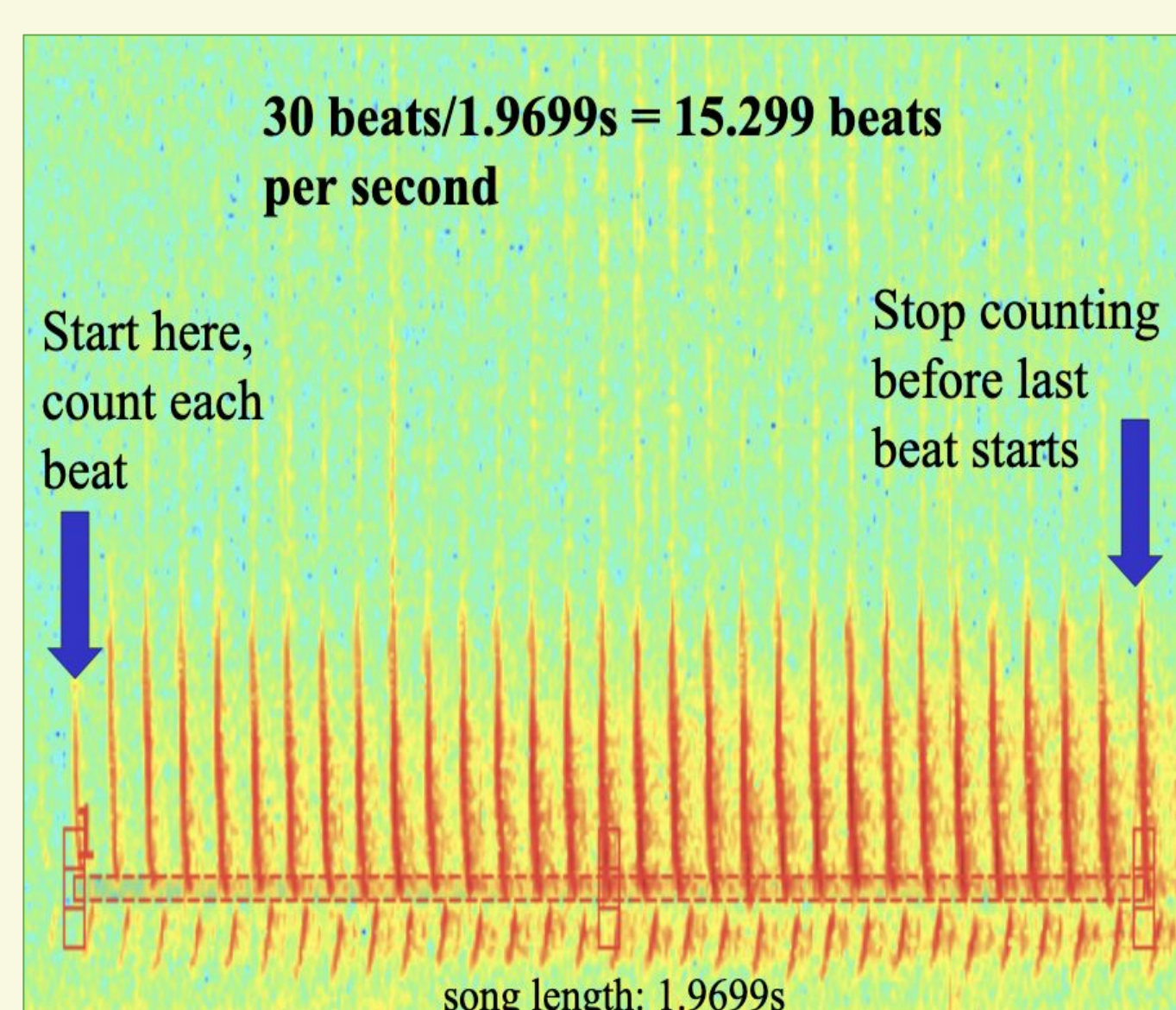


Figure 2: An example of how the speed of each song was determined in Raven.

Arrival Date vs. Song Speed

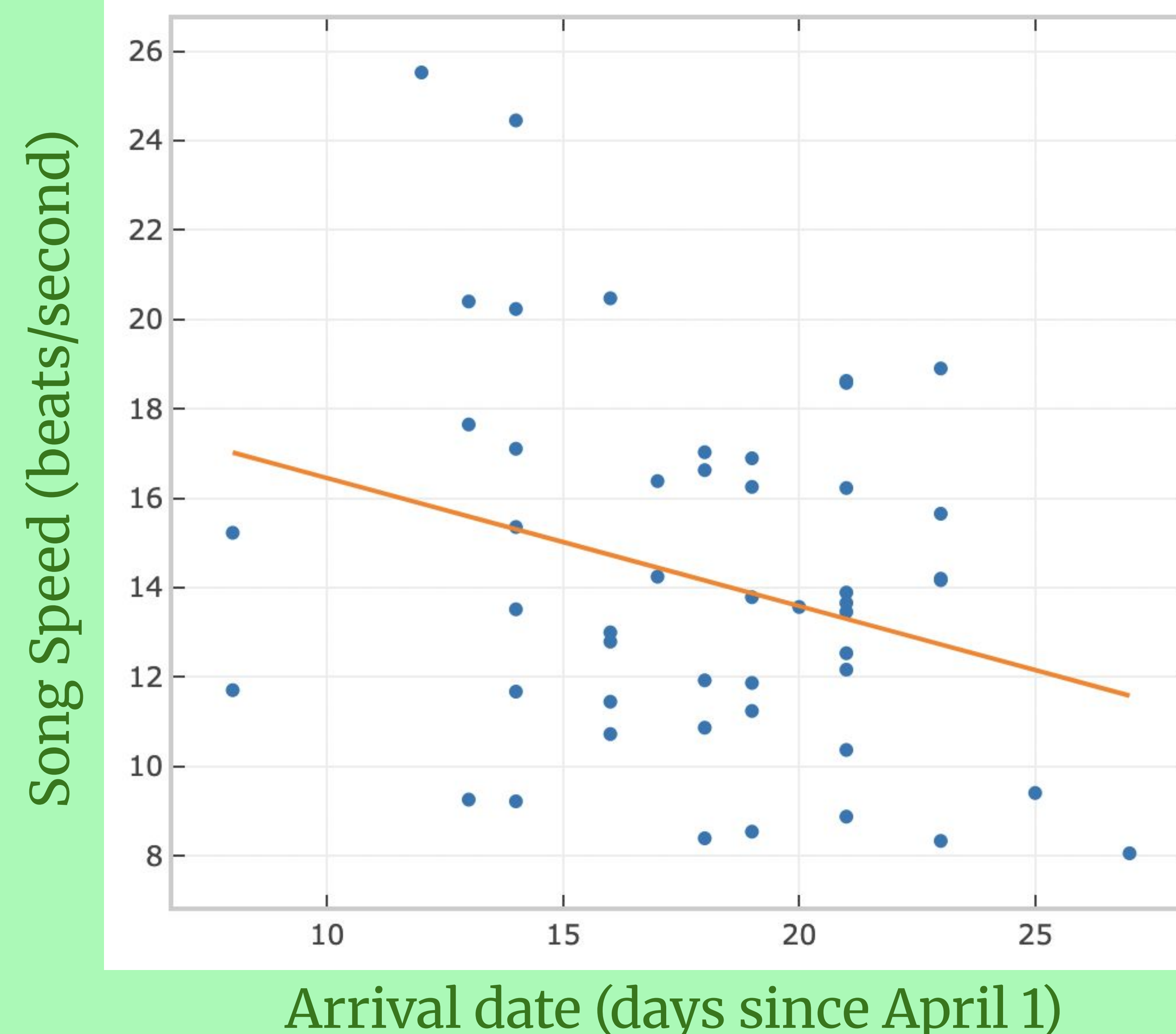


Figure 3: Graph of arrival date vs song speed; linear regression slope of -0.28631

RESULTS

- Linear regression
 - Revealed a significant correlation between arrival date and song speed among individuals identified (p-value = 0.04555)
 - Revealed a negative relationship (slope of -0.28631) between arrival date and song speed [Fig. 2]
 - As arrival date became later, song speed decreased
 - Correlation coefficient (r) = -0.28702
 - Adjusted R² = 0.06286

DISCUSSION

These data strongly support our hypothesis that Chipping Sparrows which arrive to western Massachusetts earlier also perform faster songs than those arriving later. In relation to the literature on American Redstarts, this suggests that sparrows which arrive earlier are in better physical condition, allowing them to produce a signal which has been proven to invoke a strong response in other birds.

However, American Redstarts and Chipping Sparrows are phylogenetically disparate birds, and further studies confirming whether physically fit Chipping Sparrows leave their wintering grounds earlier would further validate these findings. There are also more factors to a song's performance than speed alone, and further work on this subject will elucidate whether factors like frequency or length also correlate to arrival time.



Figure 4: Photo of a male chipping sparrow near Lederle [Isabelle Giglio 2025]

References

- [1] Goodwin SE, Podos J. Team of rivals: alliance formation in territorial songbirds is predicted by vocal signal structure. 2014 Feb
- [2] Marra PP, Hobson KA, Holmes RT. Linking winter and summer events in a migratory bird by using stable-carbon isotopes. Science. 1998 Dec

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